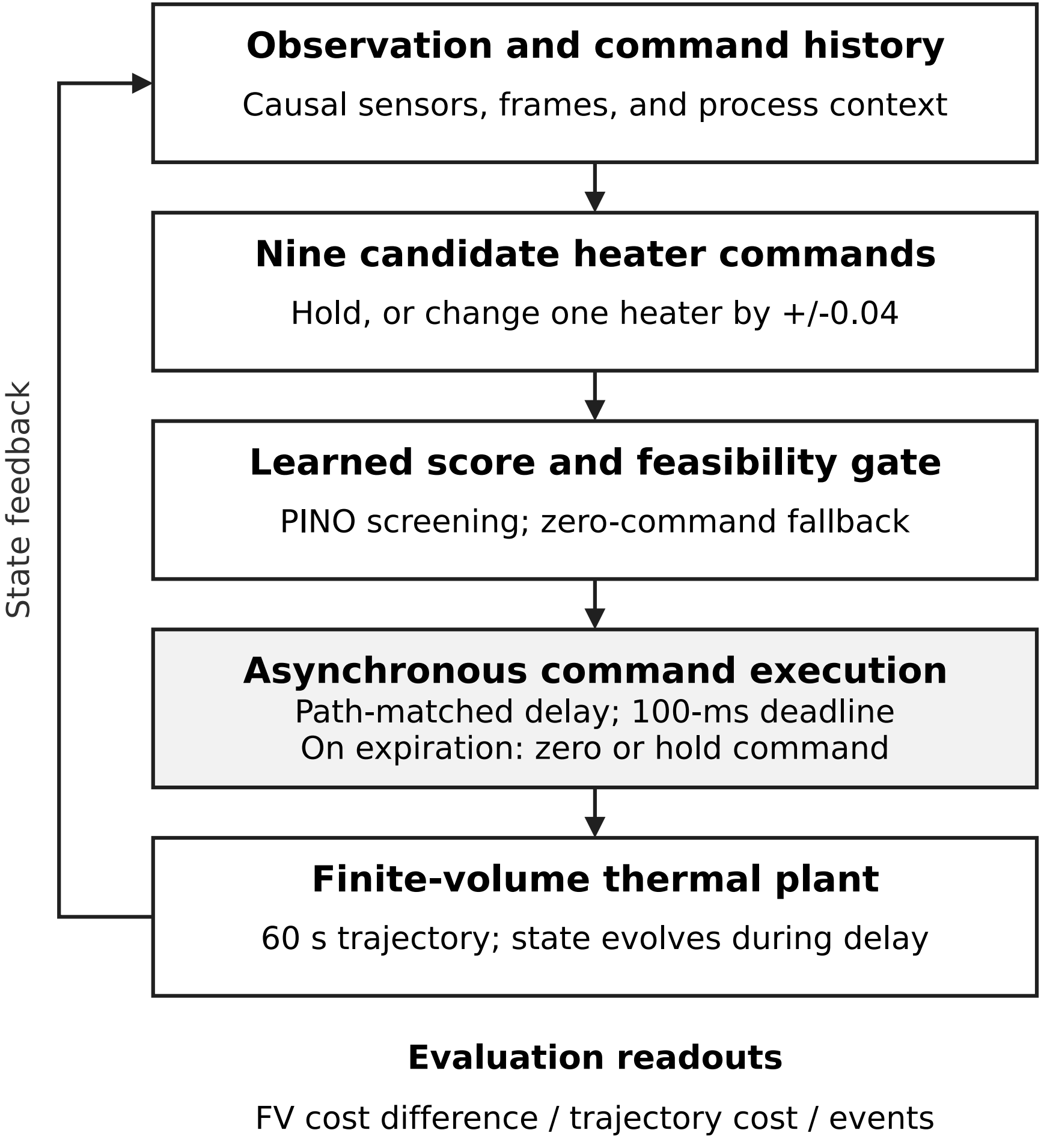
Observation and command history
Causal sensors, frames, and process context
Nine candidate heater commands
Hold, or change one heater by +/-0.04
Learned score and feasibility gate
PINO screening; zero-command fallback
Asynchronous command execution
Path-matched delay; 100-ms deadline
On expiration: zero or hold command
Finite-volume thermal plant
60 s trajectory; state evolves during delay
State feedback
Evaluation readouts
FV cost difference / trajectory cost / events